

# For A Few GB More

**Deciding upon a hard disk is a matter of knowing what exactly you want—in terms of space, speed and SATA—and about parameters such as cost per GB**

I was in Bangalore visiting some friends, and since I'm considered the "tech shopper" amongst my zany bunch of friends—mainly because I'm probably the only one who's even opened up a computer—I was immediately put to work. "Get me a hard disk," two friends demanded as soon as I walked out of the airport exit. "Hah, some holiday this is turning out to be," I thought! Both friends explained their needs while we were stuck in a routine traffic jam, though all I could think was, "Now even Bangalore is overcrowded!"

The very next day I was rushed off to a place called SP Road in Bangalore, which is the city's version of Mumbai's Lamington Road.

I was looking for two hard disks, one a budget upgrade for a friend who currently has a 20 GB IDE on an old motherboard, and the other for a music fan who wants to supplement his existing storage with something like a 250 or 300 GB SATA drive. For the first, I went around asking for an 80 GB IDE from all known brands. The prices for this size of disk ranged from Rs 2,200 for Samsung or Seagate to Rs 2,800 for Hitachi drives.

There was little to distinguish between the drives. They were all the standard 7200 rpm, and seek times were 8 - 10 ms. I was surprised to see that one shop still stocked the older 5400 rpm drives from Samsung and Seagate. It's pointless to opt for one of these—the performance hit is noticeable, and the price difference negligible. After checking out three or four shops, it came down to choosing between the 6Y080P0 Maxtor 7200 rpm 8MB buffer drive and the WD Caviar WD800JB with the same specs. The tried and tested Seagate Barracuda was available, but only had a 2 MB buffer. In the end, I decided on the WD Caviar for Rs 2600.

While asking about the IDE drive, I was also looking for SATA and SATA 2 disks. In this segment you will find 200, 250, 300, 320, 400 and even 500 GB disks, all of them 7200 rpm. Between the 160 and 300 GB mark, as capacity increases, the cost per GB decreases. This is different from the case with products such as CPUs, where the cost/performance ratio increases heavily as you opt for even moderately higher models. For example, the Seagate 200 GB SATA costs around Rs 4,000; add Rs 400, and you can get a 250 GB (25 per cent more storage for 10 per cent extra cost). The 300 GB Seagate comes for around Rs 6,000; this

gives you 50 per cent more storage than the 200 GB, for an equivalent 50 per cent increase in price. However, once you're at the 400 GB mark, things change: it costs a whopping Rs 11,000, while the 500 GB is just ridiculous at Rs 17,000!

I decided that the sweet spot was 300 GB, where the price/cost equation was more or less balanced. Now came the choice between the brands. I asked for the specifications of different models until the shopkeepers began showing signs of disinterest, and worse still, chasing me out to accommodate less curious buyers!

## SATA Or SATA 2?

There seems to be a lot of confusion between SATA and SATA 2, and a long lecture is due, but we'll save that for later. For now, let me just tell you that SATA 2 specifies certain additional features apart from speed, like being hot-swappable. You can plug it into your motherboard while the PC is On, and it will be recognised as a mass storage device, similar to a thumb drive. This, coupled with the speed boost, made me decide that SATA 2 was the way to go.

I found a shop selling 300 GB SATA 2 drives from Western Digital, Seagate, and Maxtor, with 16 MB buffers, and also a 320 GB Hitachi with 8 and 16 MB buffer models. The highest-capacity SATA 2 Samsung that was available was a 160 GB. Also available is a slightly more expensive 320 GB with 16 MB of buffer, but it wasn't available immediately (if you get it while you shop, it is worth considering).

The HDT725032VLA360 320 GB Hitachi with 8 MB of buffer and S.M.A.R.T. technology was what I decided on, and it cost me Rs 6,800 (but remember that prices will vary almost every week and from city to city).

My friend was quick to ask, why not the Western Digital 300 GB with 16 MB of buffer? I had to explain that the Hitachi's higher density per platter outweighs the buffer difference, giving better performance. When you have more data on less space, the mechanical strain caused to and by the read/write head is reduced. Thus, you also get longer life. ■



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